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Concise Report

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Organocatalytic Access to C1-Functionalized Indolizines via Remote Stereocontrolled Construction of Axially Chiral Allenes

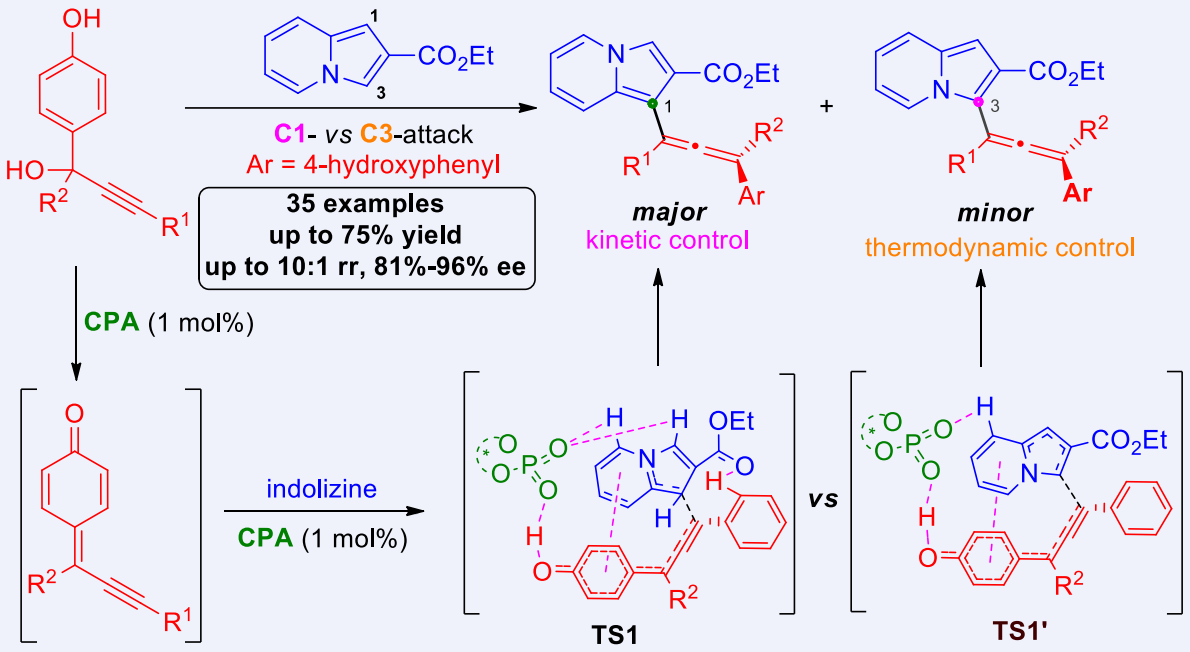
Xuan Zhang,[a] Xu-Hua Jiang,[a] Chun-Dong Huang,[b] Cheng Gao,[a] Shao-Fei Ni,*[,b,c] and Qi-Jian Ni*[,a]

[a] *College of Chemistry and Materials Science, Key Laboratory of Functionalized Molecular Solids, Ministry of Education, Anhui Laboratory of Molecule-Based Materials, Anhui Normal University, Wuhu, Anhui 241002, China*
[b] *Department of Chemistry, Shantou University, Shantou, Guangdong 515063, China*
[c] *Chemistry and Chemical Engineering Guangdong Laboratory, Shantou, Guangdong 515063, China*

Keywords

Asymmetric catalysis | Enantioselectivity | Organocatalysis | Chiral allenes | Indolizines | 1,8-Conjugate addition | Regioselectivity | Chiral phosphoric acid | Axial chirality

Comprehensive Summary



We present here a straightforward protocol for the enantioselective synthesis of axially chiral tetrasubstituted allenes enabled by chiral phosphoric acid (CPA)-catalyzed 1,8-addition of indolizines to propargylic alcohols. This strategy achieves enantioselective and regioselective C1-functionalization of indolizines under mild reaction conditions with low catalyst loading, broad functional group tolerance (35 examples), and excellent enantioselectivities (up to 96% ee). Moreover, the scale-up reaction and late-stage C3-functionalizations demonstrated its potential applications. DFT calculations were carried out to clarify the reaction mechanism and the origin of enantio- and regioselective control.

*E-mail: qijianni@ahnu.edu.cn; sfni@stu.edu.cn

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Background and Originality Content

Axially chiral allenes are important structural motifs commonly found in natural products,[1] pharmaceuticals,[2] and chiral catalysts or ligands (Figure 1a).[3] Consequently, numerous methods for the catalytic asymmetric synthesis of chiral allenes have been developed,[4] primarily including direct functionalization of allenes,[5] conversion of prochiral 1,3-enyne,[6] and isomerization of alkynes.[7] However, most of the established approaches focus on synthesizing 1,3-di- or trisubstituted chiral allenes. Very recently, organocatalytic 1,6-[8] or 1,8-conjugate addition[9] of propargylic alcohol has been demonstrated as a powerful method for the construction of chiral tetrasubstituted allenes. Despite these advances, driven by the persistent challenges in accessing chiral tetrasubstituted allenes formation, the development of new catalytic enantioselective approaches is still highly valuable and desirable.

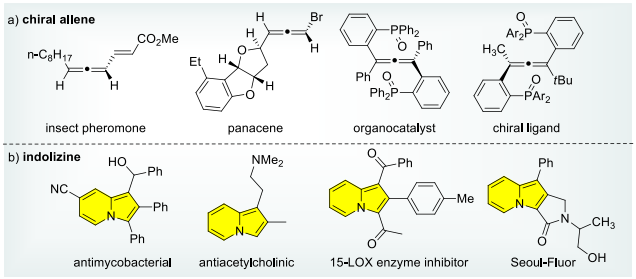
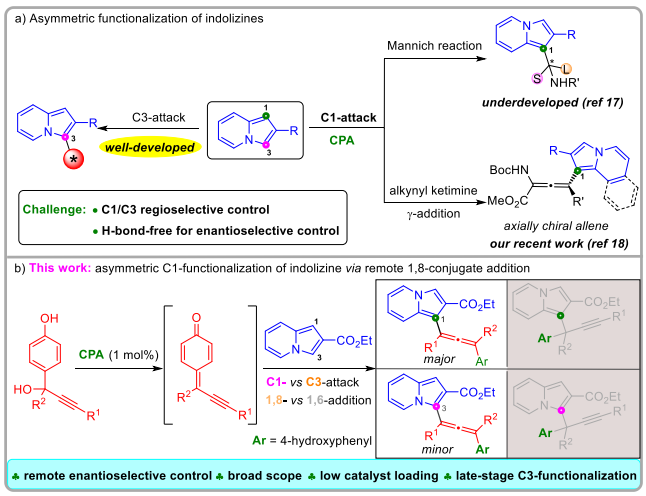


Figure 1 Representative examples of functional molecules containing chiral allene and indolizine moiety.

On the other hand, indolizines have garnered increasing attention due to their distinctive structural features and unique properties, which render them highly promising for biomedical and fluorescence applications (Figure 1b).[10] As a result, extensive efforts have been devoted to synthesizing structurally diverse indolizines over the past decades.[11] In recent years, the direct functionalization of indolizines has emerged as an efficient synthetic route for modifying indolizine moieties and attracted considerable attention. Among these, several attractive C3-functionalizations of indolizines have been reported in this field, including C–H activation,[12] propargylation,[13] Lewis acid catalysis,[14] visible-light or electro-induced,[15] as well as organocatalytic 1,4-conjugate addition (Scheme 1a).[16] In sharp contrast to the well-developed C3-functionalization of indolizine, the direct and regioselective C1-functionalization of indolizine remains challenging and elusive, especially for the C3-unsubstituted derivatives. Very recently, only very few reports concerned on the regioselective and enantioselective C1-functionalizations of indolizines through Mannich reaction.[17] Our group also successfully disclosed a CPA-catalyzed regio- and enantioselective C1-arylation of β,γ-alkynyl-α-imino esters with pyrrolo[2,1-*a*]isoquinolines.[18] Building upon these results, we became interested in this promising asymmetric C1-functionalization of indolizine via remote 1,8-conjugate addition of *in situ* generated propargylic *para*-quinone methides, providing an organocatalytic protocol for the construction of axially chiral tetrasubstituted allenes. Toward this goal, several challenging issues should be concerned: 1) controlling nucleophilic attack by indolizines at C1 *versus* C3-position; 2) achieving site-selective 1,6- or 1,8-conjugate addition of propargylic-*para*-quinone methides; 3) establishing remote stereocontrol from the reaction site without hydrogen bonding involvement.

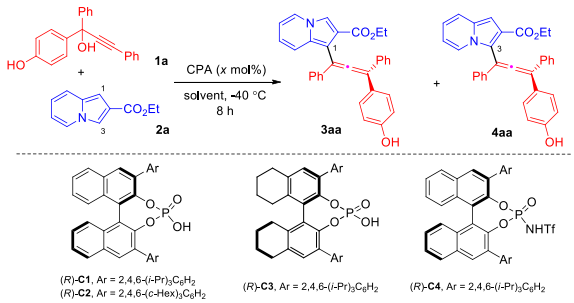
Scheme 1 Catalytic enantioselective functionalization of indolizines



Results and Discussion

To achieve this transformation, we initiated our investigation by selecting racemic propargylic alcohol **1a** and indolizine **2a** as the model substrates (Table 1). The initial test was performed in the presence of 10 mol% CPA catalyst **C1** at room temperature. To our delight, the desired chiral allene **3aa** was obtained in 46% yield with 50% ee and the regioisomeric C3-adduct **4aa** was isolated in 15% yield (entry 1). Notably, the absence of the 1,6-addition adduct confirmed the exclusive 1,8-site specificity of this approach. Decreasing the reaction temperature to –40 °C improved the ee value to 75% while maintaining a comparable yield upon extended reaction time (entry 2). To further enhance the enantio- and regioselectivity, some other catalysts with different substituents and scaffolds were then evaluated, and catalyst **C2**, possessing bulkier 2,4,6-tricyclohexylphenyl substituents at the

Table 1 Optimization of the reaction conditions[a]



Entry	CPA	Solvent	Yield[b]/%	rr (**3aa**/**4aa**)[c]	ee[d]/%
1[e]	**C1**	DCE : *c*-hexane 4 : 1	46	3 : 1	51
2	**C1**	DCE : *c*-hexane 4 : 1	49	3 : 1	75
3	**C2**	DCE : *c*-hexane 4 : 1	73	5 : 1	93
4	**C3**	DCE : *c*-hexane 4 : 1	49	2 : 1	81
5	**C4**	DCE : *c*-hexane 4 : 1	12	1 : 1.6	69
6	**C2**	DCE : toluene 4 : 1	51	3 : 1	91
7	**C2**	DCE : CCl_4 4 : 1	49	2 : 1	93
8	**C2**	DCE : $CHCl_3$ 4 : 1	46	1 : 1	91
9	**C2**	DCE : *n*-hexane 4 : 1	67	4 : 1	93
10[f]	**C2**	DCE : *c*-hexane 4 : 1	73	5 : 1	93
11[g]	**C2**	DCE : *c*-hexane 4 : 1	73	5 : 1	93
12[h]	**C2**	DCE : *c*-hexane 4 : 1	73	5 : 1	93

[a] Reaction conditions: **1a** (0.05 mmol), **2a** (0.06 mmol) and CPA (10 mol%) in 0.5 mL solvent at –40 °C for 8 h. [b] Isolated yields of **3aa**. [c] Determined based on the masses of the isolated isomers. [d] Determined by HPLC analysis on a chiral stationary phase. [e] Performed at room temperature. [f] 5 mol% CPA. [g] 2.5 mol% CPA. [h] 1 mol% CPA for 18 h.

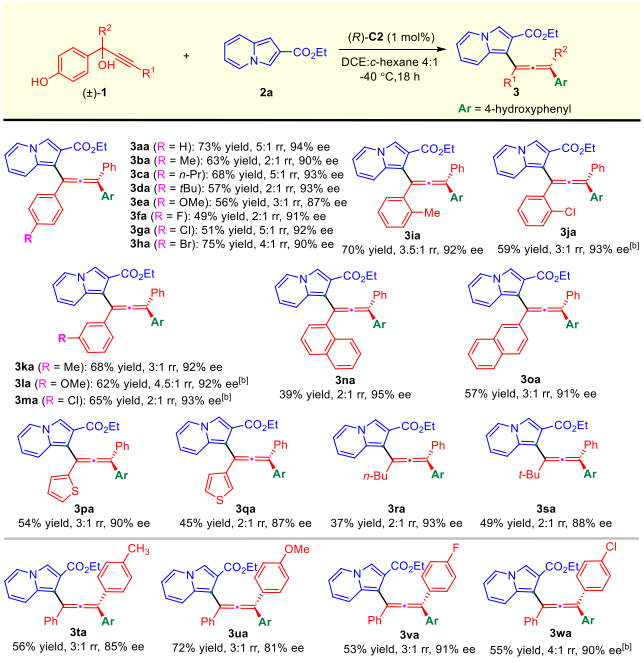
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3,3′-positions, proved optimal in terms of reactivity, regioselectivity and enantioselectivity (entry 3). Encouraged by these promising results, we screened various mixed solvents. No superior alternatives were identified when mixing DCM with other solvents, such as toluene, CCl_4, $CHCl_3$, or *n*-hexane (entries 6–9). Subsequently, catalyst loading was investigated due to the high reactivity (entries 10–12). Notably, the enantio- and regioselectivity of chiral allene **3aa** were well maintained with only 1 mol% of **C2** upon extended reaction time (18 h, entry 12). Control experiments confirmed that no reaction occurred in the absence of catalyst.

With the optimal reaction conditions established, we next explored the scope and generality of the reaction. As summarized in Scheme 2, this 1,8-addition proceeded efficiently with a wide range of propargylic alcohols **1**. Various substituents on the phenyl-tethered alkyne (R^1) were well tolerated, affording the C1-selective 1,8-adducts **3aa**–**3ma** in moderate to good yields, excellent enantioselectivities and regioselective ratios (rr) of 2 : 1 to 5 : 1. The electronic nature and the position of the substituents on the aromatic R^1 moiety did not significantly affect the formation of tetrasubstituted chiral allenes. Furthermore, naphthyl and heterocyclic substituents also reacted smoothly with indolizine **2a**, producing the corresponding **3na**–**3qa** in moderate yields, acceptable rr and 87%–95% ee. In addition, alkyl (R^1 = *n*-Bu) substituted propargylic alcohols furnished the chiral allene **3ra** in 37% yield with 93% ee and 2 : 1 rr. Remarkably, the more challenging bulky *t*-butyl group on the alkyne was compatible, delivering **3ra** with only a slight decrease in enantioselectivity. The propargylic aryl ring (R^2) was also amenable to this transformation. Both electron-donating (Me or MeO) and electron-withdrawing (F or Cl) substituents on the phenyl ring of R^2 were compatible and gave the corresponding chiral allenes **3ta**–**3wa** in 53%–72% yields, 81%–91% ee and rr of 3 : 1–4 : 1.

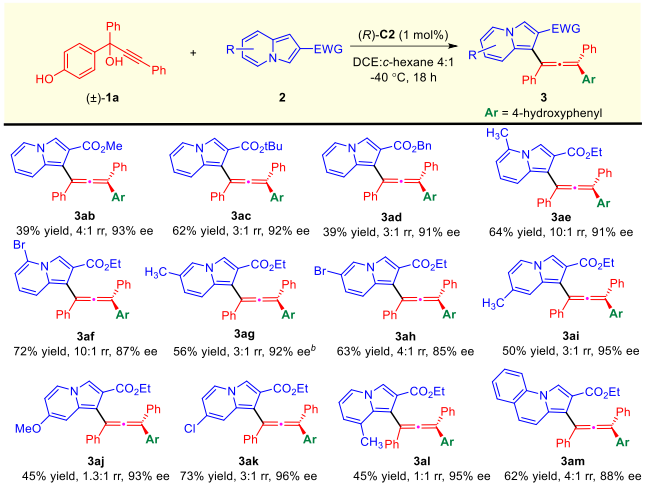
Scheme 2 Scope of propargylic alcohols **1**[a]



Unless noted otherwise, all reactions were carried out with conditions as stated in Table 1, entry 12 on 0.1 mmol scale. All yields of **3** are isolated yields after purification by column chromatography. The regioisomers were separated. The ee values were determined by HPLC analysis on a chiral stationary phase. [b] With 5 mol% of **C2**.

The scope of the indolizines was then evaluated by the 1,8-addition with propargylic alcohols **1a** (Scheme 3). Variations on the ester moiety at the C2 position of indolizine were well tolerated under the optimal conditions and furnished the desired **3ab**–**3ad** in moderate yields with excellent enantioselectivities (91%–93% ee) and good regioselective ratios (rr 3 : 1–4 : 1). Substituents on the indolizine ring were also investigated. As expected, the reaction with indolizines bearing substituents at C5 position blocked the 1,8-addition at C3 site, thus affording adducts **3ae** and **3af** in excellent regioselectivities, good yields and 87%–91% ee. Furthermore, both electron-donating and electron-withdrawing groups at C6 and C7 positions of indolizines proceeded smoothly to produce **3ag**–**3ak** in 45%–73% yields with 85%–96% ee. In addition, 8-methylindolizine proved a suitable substrate, affording **3al** in 45% yield with 95% ee and 1 : 1 rr. Significantly, extension of this protocol to the pyrroloquinoline ring successfully produced the expected **3am** in 62% yield with excellent enantio- and regioselectivity (88% ee, 4 : 1 rr), demonstrating broad compatibility of our methodology.

Scheme 3 Scope of indolizines **2**[a]



[a] All reactions were carried out with conditions as stated in Table 1, entry 12 on 0.1 mmol scale. All yields of **3** are isolated yields after purification by column chromatography. The regioisomers were separated. The ee values were determined by HPLC analysis on a chiral stationary phase. [b] With 5 mol% of **C2**.

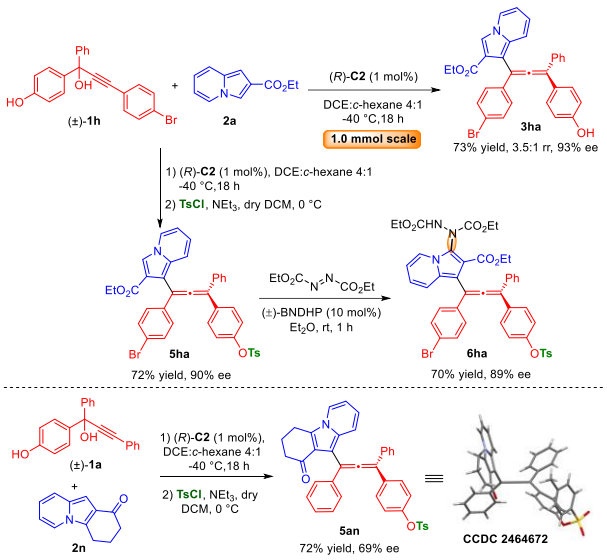
In order to demonstrate the synthetic utility of this protocol, scale-up preparation and product derivatization were conducted (Scheme 4). A scale-up reaction between racemic **1h** and **2a** proceeded smoothly, giving product **3ha** in 73% yield with 93% ee and 3.5 : 1 rr. To further showcase the synthetic value of this work, we first protected the hydroxy group via a one-pot reaction with TsCl, yielding sulfonate **5ha** in 72% yield and 90% ee. Notably, the unsubstituted C3-position provided great potential for further functionalization. For example, the direct C–H amination of **5ha** with azodicarboxylate proceeded smoothly to yield product **6ha** in 70% yield with 89% ee. In addition, subjecting C3-substituted indolizine **2n** to the one-pot reaction also delivered sulfonate **5an** in 72% yield and 69% ee. The structure of product **5an** was elucidated by X-ray crystallographic analysis, and the absolute configuration of product **5ha** was unambiguously assigned by comparison between its experimental and calculated electronic circular dichroism spectra, which was confirmed to be the (*S*) configuration (for details, see the SI).

To elucidate the origins of the excellent stereocontrol and regiocontrol in this catalytic system, several control experiments were performed. Introducing a secondary amide group at C2-position of the indolizine **2p** resulted in only 9% conversion with 40% ee value of **3ap**, indicating a detrimental H-bond interaction between the indolizine and the CPA catalyst (Scheme 5a). Furthermore, considering the detection of accumulated intermediate **QM**, we isolated **QM** in the presence of Brønsted acid and conducted

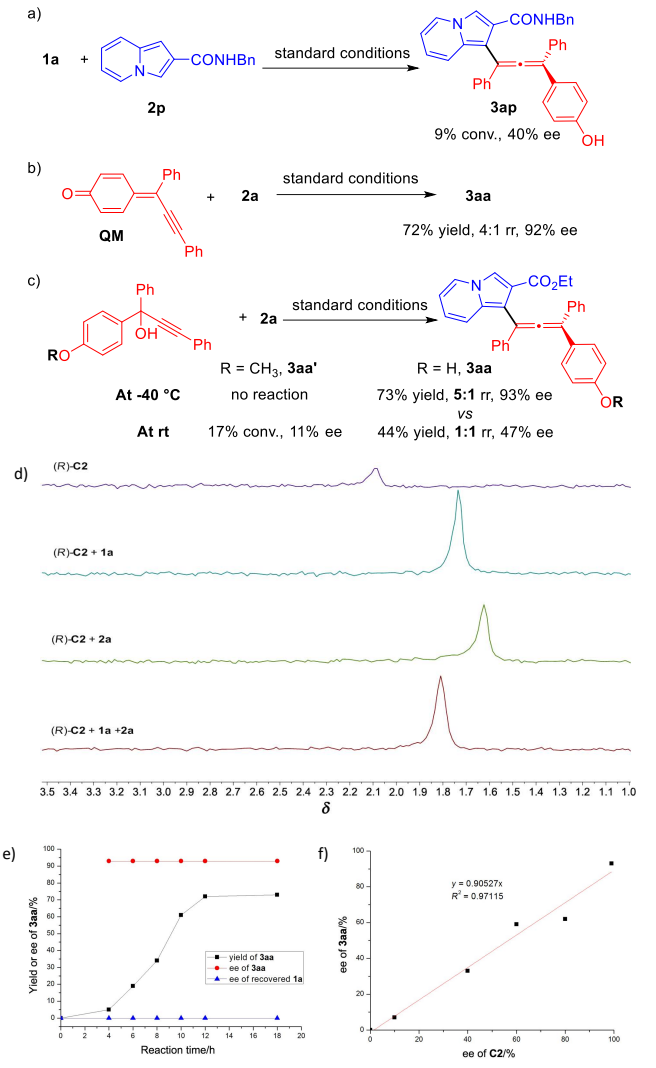
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Scheme 4 Scale-up synthesis and synthetic transformation of **3ha**



Scheme 5 Mechanism study. a–c) Control experiments. d) ^{31}P NMR study. e) Time-course study. f) Nonlinear effects



the reaction with indolizine **2a** under the standard conditions. The reaction proceeded smoothly, affording **3aa** with comparable results (Scheme 5b). This observation in combination with the control experiment that no reaction occurred between methylated propargylic alcohol **1a'** and indolizine **2a** under the standard reaction conditions demonstrated that 1,8-addition was the rate- and enantio-determining step and ruled out an ion-pairing activation mode. The kinetic studies as shown in Scheme 5e indicated that ee value of the product remained constant (93% ee) and substrate **1a** remained racemic throughout the reaction progress. These results, combined with the related reports, implied the enantioselective outcome relied entirely on the CPA catalyst via a dynamic kinetic resolution during the catalytic process.[9a-b] In addition, several control ^{31}P NMR experiments disclosed the weak interaction of CPA catalyst with propargylic alcohol **1a** and indolizine **2a** (Scheme 5d). Interestingly, the reaction performed at –40 °C in the presence of catalyst **C2** delivered C1-functionalized **3aa** in 93% ee with 5 : 1 rr. Upon warming the mixture to room temperature, the **3aa** : **4aa** ratio dropped to an equilibrium value of 1 : 1 (Scheme 5c). This finding establishes **3aa** as the kinetically controlled C1-regioisomer, whereas elevated temperatures drive equilibration toward the thermodynamically favored C3-regioisomer **4aa**. Finally, the observed linear proportionality between the ee values of product **3aa** and catalyst **C2** (Scheme 5f) strongly supported a monomolecular catalyst involvement in the enantio-determining 1,8-addition transition state.

To gain deeper mechanistic insights, density functional theory (DFT) calculations were systematically performed. The reaction pathway begins with the dehydration of substrate **1a** catalyzed by chiral phosphonic acid (CPA), leading to the formation of key intermediate **I1**. Subsequently, substrate **2a** approaches **I1** under the stereocontrol of CPA, where two competitive pathways emerge: (1) the kinetically favored 1,8-addition proceeding through transition state **TS1** with a Gibbs free energy barrier of 15.1 kcal/mol to afford product **3aa**; and (2) the thermodynamically controlled 1,6-addition via transition state **TS1'** with a higher barrier of 18.5 kcal/mol, yielding regioisomer **4aa**. Notably, an enantioselective 1,8-addition pathway was also identified, exhibiting an energy barrier of 18.5 kcal/mol for **TS1''**. Both side reactions demonstrate a significant 3.4 kcal/mol higher energy barrier compared to the main 1,8-addition pathway, rationalizing the observed kinetic preference for **3aa** formation. To further analyze the mechanism, density functional theory (DFT) calculations were carried out. Initially, substrate **1a** was dehydrated with the chiral phosphonic acid catalyst **CPA**, resulting in the formation of intermediate **I1**. Subsequently, substrate **2a** approaches **I1** under the effect of **CPA**, undergoing 1,8-addition, overcoming a Gibbs free energy barrier of 15.1 kcal/mol to generate the kinetically controlled product **3aa**. Simultaneously, a regioselective competitive reaction occurs, producing the thermodynamically controlled product **4aa**, with a Gibbs free energy barrier of 18.5 kcal/mol for the transition states **TS1'**. Additionally, enantioselective 1,8-addition is also possible, with an energy barrier of 18.5 kcal/mol. The free energy barriers of both side reactions are 3.4 kcal/mol higher than that of the main reactions.

To elucidate the origin of regioselectivity and enantioselectivity in the reaction, three distinct transition states (**TS1**, **TS1'** and **TS1''**, Figure 2) were identified, respectively forming the main product, regioselective product and enantiomeric product. For regioselectivity, energy decomposition analysis revealed that the energy difference between **TS1** and **TS1'** primarily originates from differential catalyst-substrate interactions. Detailed structural analysis demonstrated significant differences in non-covalent interactions: **TS1** features multiple stabilizing C–H···O interactions (b1 = 2.122 Å, b2 = 2.071 Å, b3 = 2.328 Å) and π–π interaction (b4 = 3.183 Å) between the catalyst and substrate fragment, while **TS1'** possesses only one C–H···O interaction (b5 = 1.849 Å) and one π–π interaction (b6 = 3.258 Å). This disparity in stabilizing interactions accounts for the relative preference for **TS1**. All these interactions could be visualized from NCI plot (see details in Supporting Information). For enantioselectivity, energy decomposi-

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tion analysis has shown that catalyst distortion is the primary source of the energy difference between **TS1** and **TS1''**. Comparative analysis of dihedral angles revealed that the transformation from pre-intermediate **I2** (∠abcd = 168.5°) to **TS1''** involves significant structural distortion (∠a''b''c''d'' = −66.6°), whereas the corresponding angle in **TS1** only exhibits minor changes (∠a'b'c'd' = −163.9°). These computational findings provide a coherent explanation for the observed regioselectivity and enantioselectivity, as the higher energy barrier for **TS1'** and **TS1''** effectively suppresses the competing pathway, in excellent agreement with experimental results.

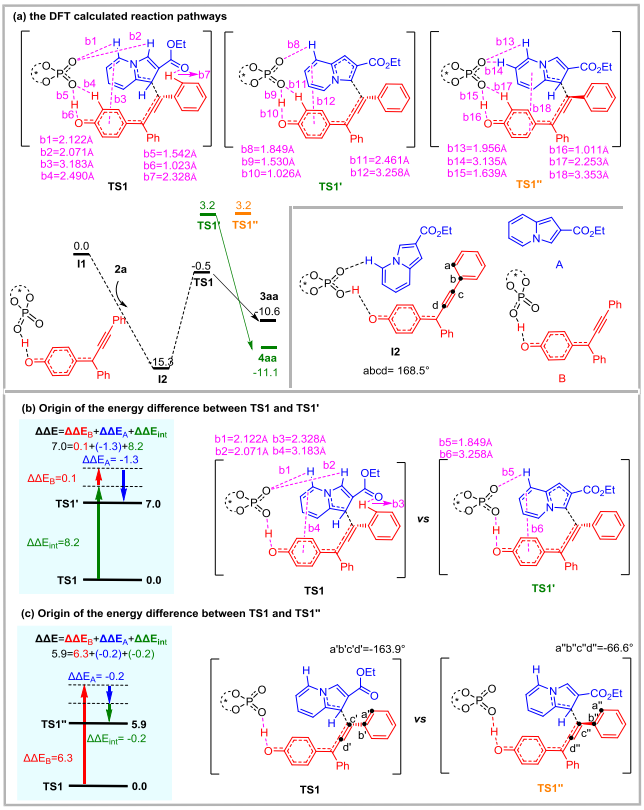


Figure 2 (a) DFT calculated reaction pathway in kcal/mol; (b) and (c) the energy decomposition analysis and structure analysis on the key transition states leading to regioselective product and enantiomeric product.

Conclusions

In conclusion, we have successfully established a CPA-catalyzed 1,8-conjugate addition of propargylic alcohols with indolizines, enabling direct access to axially chiral tetrasubstituted allenes via remote stereocontrolled C1-functionalization. This strategy addresses the long-standing challenge of regioselective C1-attack on indolizines, featuring mild conditions, low catalyst loading, broad substrate scope with high regio- and enantiocontrol. The reaction scalability and late-stage derivatization demonstrated their potential as versatile chiral building blocks. Mechanistic studies and DFT calculations confirm a kinetically favored C1-attack pathway, providing insights for future asymmetric design.

Experimental

General experimental procedures: To a 4 mL vial was added propargyl alcohol **1** (0.1 mmol, 1.0 equiv), indolizine **2** (0.12 mmol, 1.2 equiv) and (*R*)-**C2** (0.001 mmol, 1 mol%) in DCE : *c*-hexane = 4 : 1 (1 mL, *V*/*V*). The mixture was stirred at −40 °C and monitored by TLC until completion of the reaction. The reaction mixture was directly subjected to flash column chromatography on silica gel (petroleum ether/ethyl acetate 8 : 1) to afford the product **3**.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.70188.

Acknowledgement

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The Authors



Left to Right: Xuan Zhang, Xu-Hua Jiang, Chun-Dong Huang, Cheng Gao, Shao-Fei Ni and Qi-Jian Ni

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